

AI Capstone Project Idea Proposal

1. Project Title

Catchy Title: Dev-Mentor **Descriptive Name:** An Agentic RAG-Based System for Automated Context-Aware Code Review and Junior Developer Mentorship.

2. Project Idea (Problem you're solving?)

What is the problem happening?

The Egyptian software industry is facing a "Seniority Bottleneck." As the country aggressively scales its IT offshoring sector, there is a massive influx of junior developers and trainees (over 200,000 annually), while the number of senior architects capable of mentoring and reviewing code remains limited. Currently, senior developers in Egyptian software houses spend **30% to 40% of their weekly capacity** on manual code reviews and fixing repetitive "junior-level" syntax or logic errors. This manual process leads to:

- **Production Delays:** High-value projects are stalled waiting for senior approval.
- **Technical Debt:** Under pressure to meet deadlines, teams often skip thorough reviews, leading to "software erosion" that costs companies significantly in future maintenance.
- **Senior Burnout:** High-paid talent is used for "linting-style" tasks rather than high-level architecture.

Who experiences it?

- **Primary Users:** The **240+ offshoring companies** and 270+ global service delivery centers currently operating in Egypt (e.g., in Maadi Tech Park, Giza, and Alexandria).
- **Secondary Users:** The **215,000 new tech workers** projected to join the workforce by 2026 under the "Digital Egypt Strategy," who currently lack immediate, real-time feedback on their code quality.

Why is it worth solving? (Benefits & Cost Reduction)

- **Protecting Export Targets:** Egypt aims to increase digital exports to **\$9 billion by 2026**. Maintaining international quality standards (ISO/IEC) is critical to competing with Eastern Europe and India.
 - **Cost Efficiency:** Egyptian offshoring is currently **20-30% more cost-competitive** than Eastern Europe. Automating the initial 80% of code reviews using AI (Dev-Mentor) preserves this cost advantage by allowing senior developers—who earn up to **EGP 1.2M+ annually**—to focus exclusively on complex business logic rather than basic code quality.
 - **Accelerating ROI:** By reducing the "Review-Feedback-Fix" cycle from days to minutes, Egyptian software houses can increase their project throughput by an estimated **15-20%**, directly impacting the sector's contribution to the GDP (currently 5%, targeting 8%).
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3. Users

- **Primary Users:** Junior to Mid-level Software Engineers who need immediate, educational feedback on their code.
 - **Secondary Users:** Senior Developers and Tech Leads who use the system to define "Best Practice" rules and monitor team progress.
 - **Technicality:** Users are highly technical.
 - **Geographic Focus & Language:** Targeted at the **Egyptian** tech market with support for **English** (as code and technical docs are primarily in English).
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4. Proposed Solution (High-level)

Dev-Mentor is an **intelligent assistant and developer tool** integrated directly into the software development lifecycle.

Main Flow: When a developer pushes code to a repository, the system automatically intercepts the code and compares it against a knowledge base of official documentation and company-specific coding standards. The user interacts with the system via comments on their code and a specialized chat interface where they can ask for the "reasoning" behind a suggested fix.

5. Role of AI

5.1 Why is AI needed here?

A standard CRUD or rule-based application cannot understand the **intent** behind a piece of code or recognize complex patterns that vary across different projects. AI is required to provide "semantic" reviews—understanding not just *if* the code runs, but if it is *efficient and follows logic*.

5.2 What will the AI actually do?

1. **Retrieval (RAG):** The AI will search through a vector database of project documentation and style guides to find the specific rule relevant to a code snippet.
 2. **Reasoning (Agent):** It will analyze the developer's code to find violations and "reason" through a multi-step fix.
 3. **Self-Correction:** The AI will generate a refactored version of the code and attempt to verify it against existing project tests before presenting it to the user.
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6. Project Scope (Boundaries)

6.1 What the project WILL do:

- Analyze code changes in real-time.
- Provide natural language explanations for why a change is needed based on retrieved documentation.
- Suggest specific code refactors for identified issues.
- Allow developers to "chat" with their code to understand complex errors.

6.2 What the project will NOT do:

- It will not automatically deploy code to production servers.
 - It will not write entire new features or business logic from scratch.
 - It will not replace the final human sign-off required for a code merge.
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7. Data

- **Content Type:** Unstructured data including coding style guides (PDFs), internal documentation (.md files), and public software best practices.
- **Data Stored:** We will store vector embeddings of the documentation and anonymized logs of previous code reviews to improve future feedback.
- **Source:** Publicly available GitHub repositories and synthetic project documentation will be used for training and testing.

8. Team Responsibilities

Area	Responsibility
AI (Model & Intelligence)	Defining agent behavior, RAG pipeline architecture, and validating code-suggestion accuracy.
Backend & AI Integration	Managing APIs, GitHub webhook integration, and connecting the AI agents to the database.
Frontend & UI	Building the web dashboard for reports and the interface for interacting with review comments.
Testing & QA	Creating test suites to ensure the AI's "self-healed" code doesn't introduce new bugs.

Would you like me to help you refine the "Problem Statement" with specific Egyptian market data to make it even more compelling for your professors?